

CLASSIFICATION

CLASSIFICATION TECHNIQUES – PART IV

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CLASSIFICATION TECHNIQUES

The following concepts will be introduced:

- ✓ **CONDITIONAL INDEPENDENCE**
- ✓ **CLASS CONDITIONAL PROBABILITY DENSITY**
- ✓ **NAÏVE BAYES**

A **CLASSIFICATION TECHNIQUE (CLASSIFIER)** is a systematic approach to building **CLASSIFICATION MODELS** from a **DATA SET**.

CLASSIFICATION TECHNIQUES can be grouped into:

HEURISTIC; simply yet widely used like DECISION TREES, RANDOM FORESTS, NEAREST NEIGHBOR, ...

REGRESSION BASED; use a parametric conditional probability, LOGISTIC REGRESSION

SEPARATION; partition the attributes' space, SUPPORT VECTOR MACHINES and ARTIFICIAL NEURAL NETWORKS

PROBABILISTIC; exploit Bayes formula and compute posterior probability to classify records, NAÏVE BAYES, TREE-AUGMENTED NAÏVE BAYES, ...

We provide a brief overview of some of this techniques even if a formal and detailed treatment of them is out of the scope of this course.

Excellent courses on these techniques are available in Machine Learning MOOC of Andrew Ng.

A **PROBABILISTIC CLASSIFIER** solves the classification problem by computing the following **CONDITIONAL PROBABILITY**

$$P(Y | \underline{X})$$

This probability is computed exploiting the **BAYES THEOREM** or **BAYES FORMULA**

$$P(Y | \underline{X}) = \frac{P(\underline{X} | Y) \cdot P(Y)}{P(\underline{X})}$$

where

$P(Y)$ **PRIOR PROBABILITY** of the **CLASS ATTRIBUTE**

$P(\underline{X} | Y)$ **LIKELIHOOD** of the **EXPLANATORY ATTRIBUTES VECTOR** given the **CLASS ATTRIBUTE**

$P(\underline{X})$ **PROBABILITY OF THE EVIDENCE**

$P(Y | \underline{X})$ **POSTERIOR PROBABILITY** of the **CLASS ATTRIBUTE** given the **EXPLANATORY ATTRIBUTES VECTOR**

Assume the following

Y is a **BINARY CLASS ATTRIBUTE** $\{-1, +1\}$

$\underline{X}$ is a **BINARY EXPLANATORY ATTRIBUTE** $\{\text{MALE}, \text{FEMALE}\}$

Furthermore, assume the following

$$P(Y) = (0.3, 0.7) \qquad P(\underline{X} \mid Y = -1) = (0.2, 0.8) \qquad P(\underline{X} \mid Y = +1) = (0.9, 0.1)$$

Assume a record with $\underline{X} = \text{MALE}$ must be classified using the **BAYES FORMULA**

$$P(Y = -1 \mid \underline{X} = \text{Male})$$

Assume the following

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$$P(Y = -1 \mid \underline{X} = \text{Male}) = \frac{P(\underline{X} = \text{Male} \mid Y = -1) \cdot P(Y = -1)}{P(\underline{X} = \text{Male})}$$

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$$P(Y = +1 \mid \underline{X} = \text{Male}) = \frac{P(\underline{X} = \text{Male} \mid Y = +1) \cdot P(Y = +1)}{P(\underline{X} = \text{Male})} = \frac{0.9 \cdot 0.7}{P(\underline{X} = \text{Male})} = \frac{0.63}{P(\underline{X} = \text{Male})}$$

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$$Y = +1$$

What happens when the number of Explanatory Attributes grows?

If we assume that «**n**» **EXPLANATORY ATTRIBUTES** are available and all of them are **BINARY** as well as the **CLASS ATTRIBUTE** is **BINARY** then we need to know

$$\theta_{ki} = P(\underline{X} = x_k \mid Y = y_i) \quad k \in \{1, \dots, 2^n\}, \quad y_i \in \{-1, +1\}$$

n	# parameters
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1	2
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2	4
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3	8
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10	1,024
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20	1,048,576
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30	1,073,741,824
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Too many parameters to be estimated



CONDITIONAL INDEPENDENCE

ASSUMPTION

Given X , Y and Z , attributes we say that X is **CONDITIONALLY INDEPENDENT** from Y given Z , if and only if the probability of X is independent from the value of the attribute Y once the value of the attribute Z is known, formally

$$\forall i, j, k \quad P(X = x_i \mid Y = y_j, Z = z_k) = P(X = x_i \mid Z = z_k)$$

CONDITIONAL INDEPENDENCE OF EXPLANATORY ATTRIBUTES $X_1, X_2, \dots, X_n$ GIVEN THE CLASS ATTRIBUTE Y allows to drastically reduce the number of parameters to be computed; the **NAIVE BAYES ASSUMES CONDITIONAL INDEPENDENCE** and thus allows to compute the **POSTERIOR PROBABILITY OF THE CLASS ATTRIBUTE GIVEN THE EXPLANATORY ATTRIBUTES** as follows:

$$P(X_1, \dots, X_n \mid Y) = \prod_{i=1}^n P(X_i \mid Y)$$

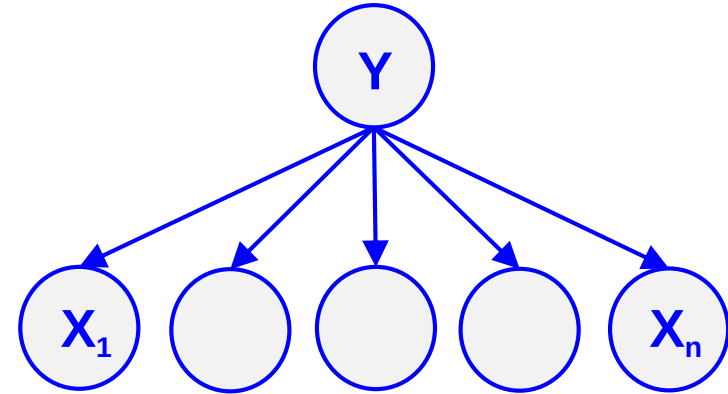
$$2 \cdot (2^n - 1) \rightarrow 2 \cdot n$$

The **NAIVE BAYES CLASSIFIER** computes the **POSTERIOR PROBABILITY OF THE CLASS ATTRIBUTE** as follows

$$P(Y = y_k | X_1, \dots, X_n) = \frac{P(Y = y_k) \cdot \prod_{i=1}^n P(X_i | Y = y_k)}{\sum_j P(Y = y_j) \cdot \prod_{i=1}^n P(X_i | Y = y_j)}$$

Then the **RECORD** ($X_1, X_2, \dots, X_n$) is **LABELED** with the **CLASS VALUE WHICH MAXIMIZES THE POSTERIOR PROBABILITY**

$$\operatorname{argmax}_{y_k} P(Y = y_k | X_1, \dots, X_n)$$

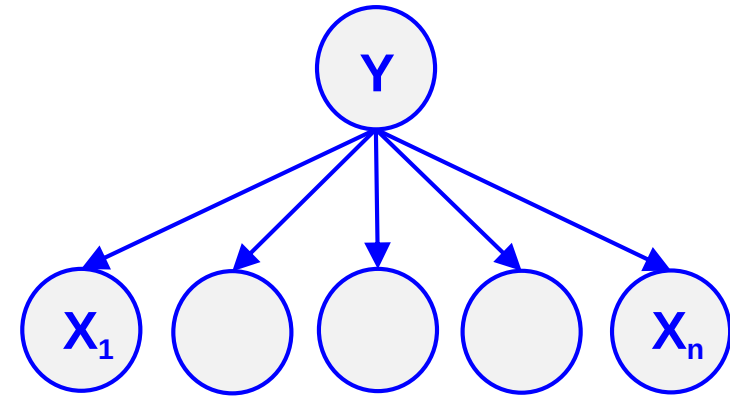


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$$\operatorname{argmax}_{y_k} P(Y = y_k | X_1, \dots, X_n)$$



Assume the following

male

female

$$P(\text{Gender} | Y = -1) = (0.2, 0.8)$$

$$P(\text{Gender} | Y = +1) = (0.7, 0.3)$$

yes

$$P(\text{Married} | Y = -1) = (0.4, 0.6)$$

no

$$P(\text{Married} | Y = +1) = (0.9, 0.1)$$

$$P(Y = -1) = 0.35 \quad P(Y = +1) = 0.65$$

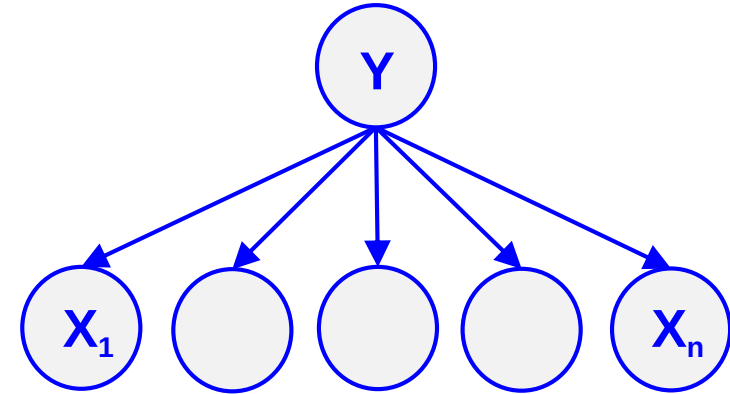
$$\underline{X} = (\text{Gender} = \text{male}, \text{Married} = \text{no})$$

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Then the **RECORD** ($X_1, X_2, \dots, X_n$) is **LABELED** with the **CLASS VALUE WHICH MAXIMIZES THE POSTERIOR PROBABILITY**

$$\arg \max_{y_k} P(Y = y_k | \text{Gender} = \text{male}, \text{Married} = \text{no})$$



Assume the following

$$P(\text{Gender} | Y = -1) = (0.2, 0.8)$$

$$P(\text{Gender} | Y = +1) = (0.7, 0.3)$$

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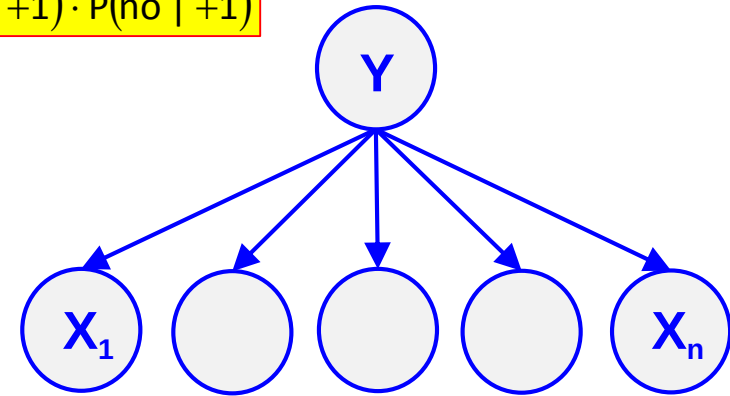
$$\underline{X} = (\text{Gender} = \text{male}, \text{Married} = \text{no})$$

$$P(-1 \mid \text{male, no}) = \frac{P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1)}{P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1) + P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1)}$$

$$P(Y = y_k \mid X_1, \dots, X_n) = \frac{P(Y = y_k) \cdot \prod_{i=1}^n P(X_i \mid Y = y_k)}{\sum_j P(Y = y_j) \cdot \prod_{i=1}^n P(X_i \mid Y = y_j)}$$

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$$\arg \max_{y_k} P(Y = y_k \mid \text{Gender} = \text{male}, \text{Married} = \text{no})$$



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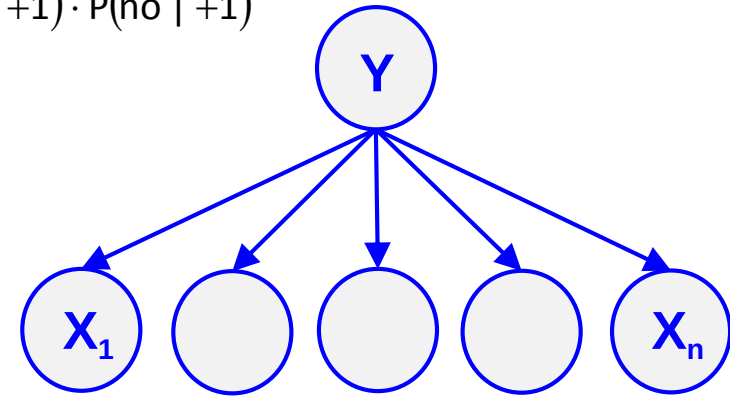
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$$P(-1 \mid \text{male, no}) = \frac{P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1)}{P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1) + P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1)}$$

$$P(-1 \mid \text{male, no}) = \frac{0.35 \cdot 0.2 \cdot 0.6}{0.35 \cdot 0.2 \cdot 0.6 + 0.65 \cdot 0.7 \cdot 0.1} = 0.48$$

Then the **RECORD** ($\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n$) is **LABELED** with the **CLASS VALUE WHICH MAXIMIZES THE POSTERIOR PROBABILITY**

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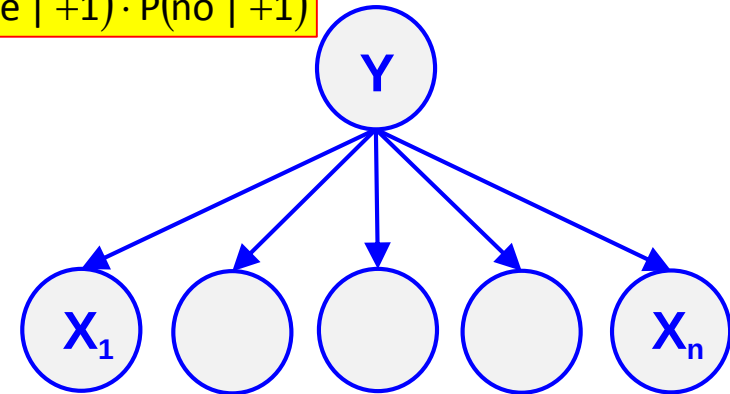
$$\underline{X} = (\text{Gender} = \text{male}, \text{Married} = \text{no})$$

$$P(+1 \mid \text{male, no}) = \frac{P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1)}{P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1) + P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1)}$$

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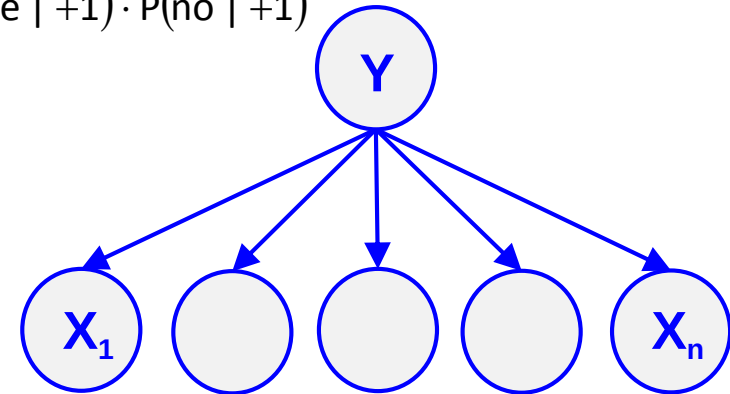
$$P(+1 \mid \text{male, no}) = \frac{P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1)}{P(+1) \cdot P(\text{male} \mid +1) \cdot P(\text{no} \mid +1) + P(-1) \cdot P(\text{male} \mid -1) \cdot P(\text{no} \mid -1)}$$

$$P(+1 \mid \text{male, no}) = \frac{0.65 \cdot 0.7 \cdot 0.1}{0.35 \cdot 0.2 \cdot 0.6 + 0.65 \cdot 0.7 \cdot 0.1} = 0.52$$

$$P(-1 \mid \text{male, no}) = \frac{0.35 \cdot 0.2 \cdot 0.6}{0.35 \cdot 0.2 \cdot 0.6 + 0.65 \cdot 0.7 \cdot 0.1} = 0.48$$

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$$\underline{\mathbf{X}} = (\text{Gender} = \text{male}, \text{Married} = \text{no})$$

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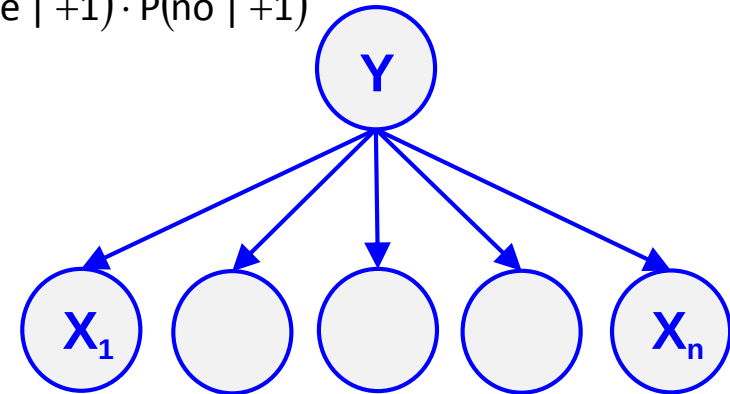
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Then the **RECORD** ($X_1, X_2, \dots, X_n$) is **LABELED** with the **CLASS VALUE WHICH MAXIMIZES THE POSTERIOR PROBABILITY**

$$\arg \max_{y_k} P(Y = y_k \mid \text{Gender} = \text{male}, \text{Married} = \text{no})$$

$$Y = +1$$



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$$P(Y = -1) = 0.35 \quad P(Y = +1) = 0.65$$

$$\underline{X} = (\text{Gender} = \text{male}, \text{Married} = \text{no})$$

The **NAIVE BAYES CLASSIFIER** just presented can be used with **CATEGORICAL ATTRIBUTES (NOMINAL and ORDINAL)**.

The **NAIVE BAYES CLASSIFIER** can also be used with **NUMERIC ATTRIBUTES (INTERVAL and RATIO)**.

Each **NUMERIC ATTRIBUTE** is associated with a **CLASS CONDITIONAL PROBABILITY DENSITY**

$$P(X_i = x_i \mid Y = y_k) = \frac{1}{\sqrt{2 \cdot \pi} \cdot \sigma_{ik}} \exp\left(-\frac{(x_i - \mu_{ik})^2}{2\sigma_{ik}^2}\right)$$

whose parameters are

$$\mu_{ik} = E[X_i \mid Y = y_k] \qquad \sigma_{ik}^2 = E[(X_i - \mu_{ik})^2 \mid Y = y_k]$$

Please note that the Naive Bayes naturally combines the use of **CATEGORICAL ATTRIBUTES (NOMINAL and ORDINAL)** with **NUMERIC ATTRIBUTES (INTERVAL and RATIO)**.